C++ Notes

What is c++ ? full explaination..

⇝ c++ is a high-Level Semi-object orienteed programming Language. developed by " Bjarne stroustrup " in the year 1979, at " BEll LAB ".

Notes :- i] c++ is a case-sensitive language.

ii] in earlier the name of c++ was " c with classes ".

iii] c++ is a portable programming Language.

Syntax :-

```
# include <iostream.h>   → Headers files
# include <conio.h>      → optional

return-type  main ()  → main function
{
      // code → std:: cout << " Hello World ! " ;
                → Statements :
}
```

comments :

1] single-line comment use [// .]

Example:

```
// Hii The example
```

2] Multi-line comment use [/*... */ .]

Example:

```
/* Hii c++
   programmer */
```

Simple example :

```
# include <iostream.h>
int main () {
    cout << " Hello world!" ;
    return 0;
}
```

→ Lessthan

output :

Hello world!

* Save [this file] using .CPP Extension and compile
→ All CPP → C++ → File

What is datatype ? full explaination.

mm>> " Defines the type's of data like (int, float, char.)

C++ has 3 type's of datatype :

1] Basic data type.
2] Derived data type.
3] user-defined datatype.

1] Basic data type :-

i) int	4 byte	-32,768 to 32768
ii) Float	4 byte	3.4E-38 to 3.4E+38
iii) char	1 byte	-128 to 127 (0 to 255)
iv) double	8 bytes	1.7E-308 to 1.7E+308.
v) bool	1 bytes	true or false
vi) void	No size	

2] derived data type :

i] Array
ii] function
iii] pointer
iv] Reference

3] user - defined data type :

i] Structure
ii] union
iii] class
iv] Enumeration

" Data types define the type of data a variable can hold".

* C++ Constants

constants are unchangeable ; when a constants variable is initialized in a program, its values cannot be changed afterwards.

- constants are also called literals. Literals

• What is Variable ? full explaination ....

⇝ Variable is the name of memory location where we store same value.

Notes : i] Variable are case- sensetive in c++.

ii] In c++, Variable must be start with either (a-z , A-Z) or _ (underscore.)

iii] we can't give extra space between the variable.

Syntax :

```
data type variable_Name ;
```

OR

```
data type variable_name = value ;
```

Variable scope :

The scope of a variable is the region in a program where the existence of that variable is valid. variable can be classified into two types :

1] Local Variable :

Local variable are declared inside the braces of any function and can be accessed only from that particular function.

2] Global Variable :

Global variable are declared outside of any function and can be accessed from anywhere.

Example :

```
# include <iostream.h>
using namespace std ;
int a=5 ; // Global variable
void func () {
```

function Definution

```
    cout << a << endl;
}
int main () {
    int a = 10;   // local variable
    cout << a << endl;
    func();  → function calling
    return 0;
}
```

output :

10
5

- What is keyword ? Full Explain ....

⇝ Key words are reserved words that have special meaning within the language. → 48+ keywords

Note : i] we can't use keyword as a variable & constant Name.

for example : int, float, if, else, while, return, for ..... etc....

- What is identifier ? full explain....

identifier is a name used to identify a variable, function class, object or any other user-define item.

Example :

```
int a;
void Mark;
class V;
```

→ identifier

- What is operators ? full explain....

Operators is symbol that is used to perform mathematical and logical ~~types~~ task.

C++ operators :-

i] Arithmetic operator (+, -, *, /, ÷)
ii] Relational operator (>, <, >=, <=, ==, !=)
iii] Logical operator (&&, ||, !)
iv] Assignment operator (=, +=, -=, *=, /=)
v] Ternary operator (? :)
vi] Bitwise operator (&, |, ^, ~, <<, >>)

- Conditional statement :-

i] if
ii] if-else
iii] else-if Ladder
iv] nested if-else
v] switch statement

- What is ~~of~~ if statement ?

——» If statement test conditions if condition is true then if block code will be executed otherwise no action taken.

Syntax :

```
if (condition)
{
    // code
}
```

- What is if-else statement ?

——» it used to execute two statement for a single condition, if given condition is true if block execute otherwise else block will be execute.

Syntax:

```
if (condition)
{
    // code - statement 1
} else {
    // code :-   "   2
}
```

- what is else-if ladder statement ?

→ else-if ladder statement is used when we have multiple condition.

Syntax:

```
if (condition)
{
    // statement 1
}
else if (condition 2)
{
    // Statement 2
}
else
{
    // statement 3
}
```

- What is nested if else statement ?

→ whenever we defined if-else block inside another if else block called nested if-else statem..

Syntax:

```
if (condition1) {
    if (condition 2) {
        Statement 1 ;
    } else {
        Statement 2 ;
    }
} else {
    Statement 3 ;
}
```

• What is switch statement ?

→ Switch statement used when we want to select only one case out of multiple cases.

Syntax:

```
Switch (exp)
{
   case 1 : statemel 1;
   break;
   case 2 : statement 2;
   break;
      :
   case n : statement n;
   break;
   default : statement;
}
```

• What is Loop ? full explain.....

→ Loops are powerful construct that help you execute code multipule times efficiently.

Types of Loops :-

1] For loop :- For loop performs all operations in single line.

Syntax :

```
For ( initialization ; condition ; increm / decre )
{
        // block of code
}
```

2] While loop :- While loop is also known as "entry controlled loop". The statement will be executed continously until the given condition is no longer satisfied.

Syntax: while (condition)
{
// statement;
// increment / Decrement;
}

3] do-while loop:

It is also known as "exit control loop", becose it tests condition at. the end of loop body.

Syntax: do {
// statement;
incren / Decre;
} while (condition);

- What is array ? Explain in Detail....

→ Array is a derived datatype which is constructed by the help of "primittive datatype".

- It stores multiple values in single variable with continous memory location.

Syntax:
data type arrayName[arraySize];

example:
int Mark[50];

- What is String ? Explain in Detail....

→ A String is a one-dimensional array of character terminated by NULL charaeters.

Syntax:-

Std :: String str; or datatype StrName[size];
→ char

- What is Structure ? Explain....

→ Structure is a user-define datatype which are used to store, dis-similar types of value

Syntax :

```
Struct    Structure_Name
{
        // member ;
};
```

- Union :- Same as a structure but one key difference: all members of a union share the same memory location.

Syntax :

```
union    union_Name
{
        // member ;
};
```

- What is Macro ? full explain....

→ Macro is a peace of code in a program which is given same name. whenwer the name is used, it is replaced by the contents of the mem.

Syntax : # define Macro-Name content

Types :-

1) object like macro
2) Function like macro.

• What is pointer ? Full Explaination .....

→ pointer are variable that store the memory address of another variable.

Note : i] & → address of operator

ii] * → value of address operators

Syntax :

datatype * var_Name ;

example :

int *P ;

• What is function ? Explain....

→ Function is a block of code that runs only when it is called.

- function is a block of code that performs a specific task.

types : i] pre- define function

ii] user- define function

i] pre- define function

1) strcpy () ;
2) str cat () ;
3) strlen () ;

ii] user- Define function

1) main ()
2) Fun ()
3) add ()

Syntax :

return type fun_Name ( ) argument

{

// your code

}